

Unit 12, Lesson 2: Finding Surface Area of Right Circular Cylinders

Benchmarks

MA.7.GR.2.1 Given a mathematical or real-world context, find the surface area of a right circular cylinder using the figure's net.

MA.7.GR.2.2 Solve real-world problems involving surface area of right circular cylinders.

Learning Targets

- ☐ I can draw the net of a right circular cylinder.
- ☐ I can find the surface area of a right circular cylinder.

12.2.1 Warm-Up: Manufacturing Engineer

Joemel Johnson, a senior manufacturing engineer, mentions that one of his responsibilities is to interpret and explain, in layman's terms (common language), the drawings of the design engineers to the technicians he works with.

1. Describe a time when you had to interpret for yourself or explain a math problem to a partner. What did you do to ensure that you or your partner understood the problem?

12.2.2 Guided Instruction: Finding the Surface Area of Right Circular Cylinder

1. Assume that a can of soda is in the shape of right circular cylinder.
- a. Complete the table to find the surface area of a can of soda whose diameter is 8 centimeters (cm.) with a height of 13 cm. Use $\frac{22}{7}$ for π . Round your answer to the nearest thousandth.

Draw a net and label its dimensions. (Include units.)	
Determine the area of one circular base. (Include units.)	
Find the sum of the areas of both circular bases. (Include units.)	
Find the lateral area. (Include units.)	
Find the sum of the areas for all the parts. (Include units.)	Sum of the Areas of the Circular Bases  Lateral Area
	$\approx$ _____

- b. What is the surface area of a can of soda whose diameter is 8 centimeters (cm.) with a height of 13 cm?
- c. Your partner is absent from class today. Write a summary describing to your partner how to find the surface area of a right circular cylinder when given the cylinder's height and diameter.
2. Evelyn stores all her gift wrap paper in a plastic storage bag that is shaped like a right cylinder. The storage bag is 41 inches (in.) tall, and its radius measures 5.6 in. What is the surface area of Evelyn's storage bag? Leave your answer in terms of π and include units.
3. Charlie calculated the surface area of a mailing tube whose diameter is 5 inches (in.) and height is 26 in. His calculation is shown. Work with your partner to determine if Charlie is correct. If not, identify his error(s) and correct his work.

$$\begin{aligned}(\pi \cdot 5^2) + 2 \cdot \pi \cdot 5 \cdot 26 \\ 25\pi + 10\pi \cdot 26 \\ 25\pi + 260\pi \\ 285\pi\end{aligned}$$



12.2.3 Guided Practice: Finding Surface Areas of Right Circular Cylinders

1. There are three different types of bass drums: tom-toms, floor toms, and kick drum. Each drum is the shape of a right circular cylinder.

Suppose you intend to place each drum individually in a heavy-duty nylon bag when traveling locally. Find the amount of nylon needed for each drum bag. Use 3.14 for π and include units in your answer.



Floor Tom

Kick Drum

Bass Drum	Work	Amount of Nylon Needed
Tom-Tom $d = 7$ in. $h = 10$ in.		
Floor Tom $d = 15$ in. $h = 16$ in.		
Kick Drum $r = 8$ in. $h = 22$ in.		

12.2.4 Your Turn!

1. A small business on Etsy sells confetti cannons for buyers to celebrate special occasions. The confetti cannons come in an enclosed tube in the shape of a right circular cylinder. Each cannon is 6 inches (in.) tall and 1.5 in. wide. How much material is used to make the enclosed tube of each cannon? Leave your answer in terms of π and include units.

12.2.5 Wrap-Up: Reflection

1. Take a **SELFIE** by checking the box that best matches what you did today.

Showed my work	☐ 	☐ 
Explained my answers	☐ 	☐ 
Lots of math vocabulary used	☐ 	☐ 
Found multiple solutions	☐ 	☐ 
I persevered through the problem	☐ 	☐ 
Eliminated careless errors	☐ 	☐ 

